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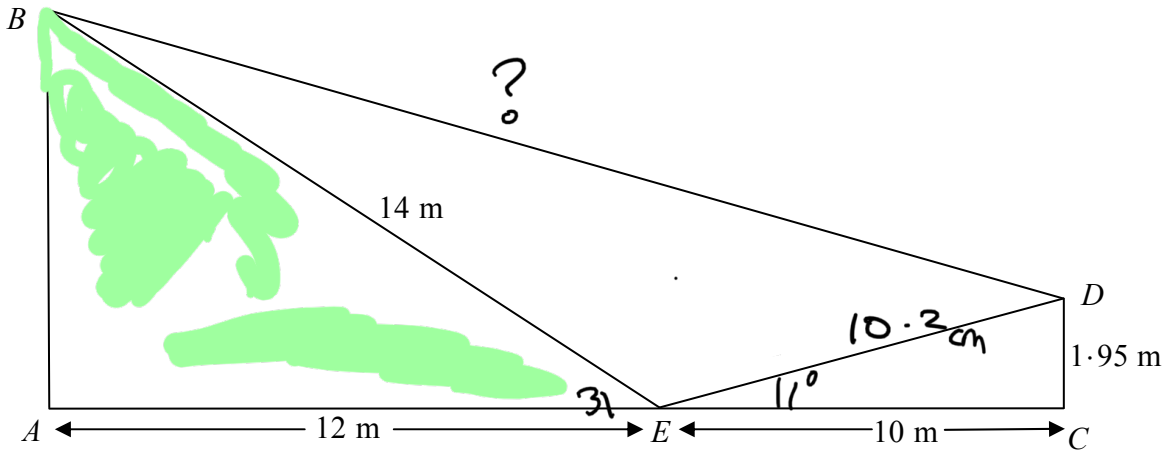
Question 9 **(30 marks)**

At an activity centre a zip-line, $[BD]$, runs between two vertical poles, $[AB]$ and $[CD]$, on level ground, as shown. The point E is on the ground, directly below the zip-line.
$|AE| = 12$ m, $|BE| = 14$ m, $|CD| = 1{\cdot}95$ m, and $|EC| = 10$ m.



(a) **(i)** Find the distance $|ED|$, correct to one decimal place.

$a^2 + b^2 = c^2$

$1{\cdot}95^2 + 10^2 = 103{\cdot}8025$

$|ED| = \sqrt{103{\cdot}8025}$

$|ED| = 10{\cdot}2\text{m}$

(ii) Find $|\angle AEB|$, correct to the nearest degree.

SOHCAHTOA

$\cos|\angle AEB| = \frac{12}{14}$

$|\angle AEB| = \cos^{-1}\frac{12}{14}$

$|\angle AEB| = 31°$

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(b) **(i)** Find $|\angle DEB|$, given that $|\angle CED| = 11°$, correct to the nearest degree.

$|\angle CED| = 11$

$|\angle AEB| + 30$

$180° - 41° = 138°$

(ii) Hence, or otherwise, find the distance $|DB|$.
Give your answer correct to one decimal place.

cosine rule

$\Rightarrow a^2 = b^2 + c^2 - 2bc\cos A$

$a^2 = 14^2 + 10.2^2 - 2(14)(10.2)\cos 138$

$a^2 = 512.28216$

$|DB| = 22.6336$